

MPECI 2025/2026 — Practical Guide 2

Exam Study Notes: Random Variables, Distributions & Inequalities

1. CORE CONCEPTS: PMF, CDF, $E[X]$, $\text{Var}[X]$

Random Variable (RV): A function that maps outcomes of a random experiment to numerical values. *Discrete* RVs take countable values; *Continuous* RVs take values over an interval.

Probability Mass Function (PMF) — discrete RVs

$$p_X(x) = P(X = x) \text{ for each value } x \text{ in the support}$$

Properties: $p_X(x) \geq 0$ for all x , and $\sum p_X(x) = 1$

Cumulative Distribution Function (CDF)

$$F_X(x) = P(X \leq x) = \text{sum of } p_X(k) \text{ for all } k \leq x$$

Properties: non-decreasing, $F(-\infty)=0$, $F(+\infty)=1$, right-continuous.

% MATLAB: `stem(x, pmf)` for PMF % `stairs(x, cdf)` for CDF (add point at left edge: `stairs([x0 x], [0 F])`)

Expected Value and Variance

Quantity	Formula (discrete)	MATLAB
$E[X]$	$\sum(x \cdot p)$	<code>sum(x .* p)</code>
$E[X^2]$	$\sum(x.^2 \cdot p)$	<code>sum(x.^2 .* p)</code>
$\text{Var}[X]$	$E[X^2] - E[X]^2$	<code>sum(x.^2.*p) - mean_x^2</code>
$\text{std}(X)$	$\sqrt{\text{Var}[X]}$	<code>sqrt(Var)</code>
$E[aX+b]$	$a \cdot E[X] + b$	— (linearity of expectation)
$\text{Var}[aX+b]$	$a^2 \cdot \text{Var}[X]$	— (constants shift, don't scale var)

■ $\text{Var}[X] = E[X^2] - E[X]^2$ is the fastest formula for calculations. Never confuse with $\text{std}[X] = \sqrt{\text{Var}[X]}$.

2. DISCRETE DISTRIBUTIONS

Uniform Discrete — die roll, $X \in \{1, \dots, n\}$

$$p(x) = 1/n \quad E[X] = (n+1)/2 \quad \text{Var}[X] = (n^2 - 1)/12$$

For a fair 6-sided die: $E[X] = 3.5$, $\text{Var}[X] = 35/12 \approx 2.917$

Binomial — $X \sim \text{Bin}(n, p)$

Models the number of successes in n independent Bernoulli trials, each with success probability p .

$$P(X=k) = C(n, k) \cdot p^k \cdot (1-p)^{(n-k)} \quad k = 0, 1, \dots, n$$

Parameter	Value	Note
$E[X]$	$n \cdot p$	4 tosses $p=0.5 \rightarrow E=2$
$\text{Var}[X]$	$n \cdot p \cdot (1-p)$	4 tosses $p=0.5 \rightarrow \text{Var}=1$
MATLAB PMF	<code>nchoosek(n,k)*p^k*(1-p)^(n-k)</code>	or use <code>binopdf(k,n,p)</code>
MATLAB CDF	<code>binocdf(k,n,p)</code>	$P(X \leq k)$

■ Binomial → Poisson when n is large, p is small, and $\lambda = n \cdot p$ is constant. The approximation is excellent for BER-style problems ($n=8000$, $p=1e-5$).

Geometric — $X \sim \text{Geo}(p)$

Models the number of trials until the **first success**. $X \in \{1, 2, 3, \dots\}$

$$P(X=k) = (1-p)^{(k-1)} \cdot p \quad k = 1, 2, 3, \dots$$

$$\text{CDF: } F(k) = 1 - (1-p)^k \quad P(X > k) = (1-p)^k$$

Quantity	Formula	With $p=0.3$
$E[X]$	$1/p$	3.333
$\text{Var}[X]$	$(1-p)/p^2$	7.778
$P(X>k)$	$(1-p)^k$	$P(X>3) = 0.7^3 = 0.343$
$P(X>3 X>1)$	$P(X>2) = (1-p)^2$	0.49 ← memoryless!

MEMORYLESS PROPERTY (Geometric & Exponential — exam classic)

$$P(X > s+t \mid X > t) = P(X > s) \text{ for all } s, t \geq 0$$

"Given you've already failed t times, the remaining distribution is identical to starting fresh." The past carries no information.

Proof: $P(X>s+t \mid X>t) = P(X>s+t)/P(X>t) = (1-p)^{(s+t)} / (1-p)^t = (1-p)^s = P(X>s) \checkmark$

✓ EXAM: $P(X>3|X>1)$ with $p=0.3$: answer = $P(X>2) = 0.7^2 = 0.49$ (NOT 0.343 which would be $P(X>3)$ unconditionally)

Poisson — $X \sim \text{Pois}(\lambda)$

Models the number of rare, independent events in a fixed interval (time, space, packets, etc.). Arises as Binomial limit.

$$P(X=k) = e^{(-\lambda)} \cdot \lambda^k / k! \quad k = 0, 1, 2, \dots$$

Quantity	Value	Key fact
$E[X]$	λ	Mean = variance (unique property of Poisson!)
$\text{Var}[X]$	λ	$\text{std}[X] = \sqrt{\lambda}$
$P(X=0)$	$e^{(-\lambda)}$	Probability of zero events
$P(X>k)$	$1 - \text{poisscdf}(k, \lambda)$	Always use complement for "more than"
Scaling	$\lambda_{\text{new}} = \lambda \cdot t$	Scale λ by time window length!

`poisspdf(k, lambda) % P(X=k) poisscdf(k, lambda) % P(X<=k) poissrnd(lambda,r,c) % random samples`

■ SCALE λ BY TIME: if rate=5/s and window=2s, use $\lambda=10$. e.g. $P(<10 \text{ msgs in } 2s) = \text{poisscdf}(9, 10)$

3. CONTINUOUS DISTRIBUTIONS

For continuous RVs, we use a **Probability Density Function (PDF)** $f(x)$. Probabilities are areas: $P(a < X < b) = \int_a^b f(x) dx$. $P(X = x) = 0$ for any single point.

Exponential — $X \sim \text{Exp}(\lambda)$

Models **waiting time** between independent events (e.g. inter-arrival times). Has the memoryless property.

$$\text{PDF: } f(x) = \lambda \cdot e^{-\lambda x} \text{ for } x \geq 0 \quad \text{CDF: } F(x) = 1 - e^{-\lambda x}$$

$$E[X] = 1/\lambda \quad \text{Var}[X] = 1/\lambda^2 \quad P(X > x) = e^{-\lambda x}$$

■ CRITICAL MATLAB TRAP — Exponential functions use MEAN, not rate

MATLAB function	Parameter	What to pass	Example $\lambda=0.2$
exppdf(x, mu)	mu = mean = $1/\lambda$	exppdf(x, 1/lambda)	exppdf(x, 5)
expcdf(x, mu)	mu = mean = $1/\lambda$	expcdf(x, 1/lambda)	expcdf(x, 5)
exprnd(mu,r,c)	mu = mean = $1/\lambda$	exprnd(1/lambda,r,c)	exprnd(5,1,N)

■ This is the #1 MATLAB mistake for exponential problems. The parameter is the MEAN ($1/\lambda$), never the rate (λ) itself.

Exponential Memoryless Property

$$P(X > s+t \mid X > t) = P(X > s) = e^{-\lambda s}$$

```
% Verify memoryless property (lambda=0.2, s=2, t=3): samples = exprnd(1/0.2, 1, 1e5); P_cond
= sum(samples > 5) / sum(samples > 3); % should equal P_Xgt2 = sum(samples > 2) / 1e5; %
P(X>2) = e^(-0.4)
```

Competition of two Exponentials

If $X \sim \text{Exp}(\lambda_X)$ and $Y \sim \text{Exp}(\lambda_Y)$, independent: $P(X < Y) = \lambda_X / (\lambda_X + \lambda_Y)$

Intuition: the process with the higher rate finishes first more often. With $\lambda_X=0.4$, $\lambda_Y=0.1$: $P(X < Y) = 0.4/0.5 = 0.8$

Gaussian (Normal) — $X \sim N(\mu, \sigma^2)$

$$\text{PDF: } f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot \exp(-(x-\mu)^2/(2\sigma^2))$$

$$E[X] = \mu \quad \text{Var}[X] = \sigma^2 \quad \text{std}[X] = \sigma$$

Rule	Probability	Interval
$\pm 1\sigma$ (68% rule)	$P(\mu-\sigma < X < \mu+\sigma) \approx 0.6827$	$[\mu-\sigma, \mu+\sigma]$
$\pm 2\sigma$ (95% rule)	$P(\mu-2\sigma < X < \mu+2\sigma) \approx 0.9545$	$[\mu-2\sigma, \mu+2\sigma]$
$\pm 3\sigma$ (99.7% rule)	$P(\mu-3\sigma < X < \mu+3\sigma) \approx 0.9973$	$[\mu-3\sigma, \mu+3\sigma]$

```
normcdf(x, mu, sigma) % P(X <= x) normcdf(b,mu,s)-normcdf(a,mu,s) % P(a < X < b) randn(r,c) %
N(0,1) samples mu + sigma*randn(r,c) % N(mu, sigma^2) samples
```

■ randn() gives $N(0,1)$. To get $N(\mu,\sigma^2)$: use $\text{mu} + \text{sigma} \cdot \text{randn}(\dots)$

4. BINOMIAL → POISSON APPROXIMATION

When n is large, p is small, and $\lambda = n \cdot p$ is **constant**, the Binomial distribution converges to $\text{Poisson}(\lambda)$.

$$\text{Bin}(n, p) \rightarrow \text{Pois}(\lambda=np) \text{ as } n \rightarrow \infty, p \rightarrow 0, np=\lambda \text{ fixed}$$

Packet size	n (bits)	$\lambda = n \cdot \text{BER}$	P(0 errors) Binomial	P(0 errors) Poisson
100 bytes	800	0.008	$(1-10^{-8})^{800} \approx 0.99202$	$e^{-(0.008)} \approx 0.99203$

1000 bytes	8000	0.08	$(1 - 10^{-4})^{8000} \approx 0.92312$	$e^{(-0.08)} \approx 0.92312$
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✓ Both methods give virtually identical answers — Poisson is much easier to compute!

5. SUM OF RVs, MARKOV, CHEBYSHEV & CLT

Sum of i.i.d. Random Variables: $S_n = X_1 + X_2 + \dots + X_n$

$$E[S_n] = n \cdot \mu \quad \text{Var}[S_n] = n \cdot \sigma^2$$

These follow directly from linearity of expectation and independence.

Markov Inequality (requires $X \geq 0$, uses mean only)

$$P(X \geq a) \leq E[X] / a \quad \text{for any } a > 0$$

Very loose bound — only valid when X is non-negative. Cookie example: $P(X_i \geq 30) \leq 20/30 = 2/3 \approx 0.667$

Chebyshev Inequality (uses mean AND variance)

$$P(|X - \mu| \geq k) \leq \sigma^2 / k^2 \quad \text{for any } k > 0$$

Tighter than Markov; works for any distribution with finite mean and variance. Cookie example: $P(|X_i - 20| \geq 10) \leq 9/100 = 0.09$

Central Limit Theorem (CLT) — for large n

$$S_n \sim N(n \cdot \mu, n \cdot \sigma^2) \quad \text{approximately, for large } n$$

$$P(a < S_n < b) \approx \text{normcdf}(b, n \cdot \mu, \sigma \cdot \sqrt{n}) - \text{normcdf}(a, n \cdot \mu, \sigma \cdot \sqrt{n})$$

Cookie $S_{100} \sim N(2000, 900)$, $\text{std}=30$: $P(1970 < S_{100} < 2030) = P(-1 < Z < 1) \approx 0.6827$ (the 68% rule)

■ The 68% rule: $P(\mu - \sigma < X < \mu + \sigma) \approx 0.6827$ — memorise this!

Method	What it uses	Bound (cookie: $P(X_i \geq 30)$)	Quality
Markov	Mean only	$\leq 20/30 = 0.667$	Very loose
Chebyshev	Mean + Variance	$P(X - 20 \geq 10) \leq 0.09$	Tighter
CLT/Normal	Full distribution	$\approx 1 - \text{normcdf}(30, 20, 3) \approx 0.0004$	Exact (large n)

6. COMPLETE DISTRIBUTION REFERENCE TABLE

Distribution	PMF / PDF	E[X]	Var[X]	MATLAB
Uniform Discrete {1..n}	$p(x)=1/n$	$(n+1)/2$	$(n^2-1)/12$	<code>ones(1,n)/n</code>
Binomial Bin(n,p)	$C(n,k)p^k(1-p)^{(n-k)}$	np	$np(1-p)$	<code>binopdf / binocdf</code>
Geometric Geo(p)	$(1-p)^{(k-1)} p$	$1/p$	$(1-p)/p^2$	$(1-p)^{(k-1)} * p$
Poisson Pois(λ)	$e^{(-\lambda)} \lambda^k / k!$	λ	λ (=mean!)	<code>poisspdf / poisscdf</code>
Exponential Exp(λ)	$\lambda e^{(-\lambda x)} \quad x \geq 0$	$1/\lambda$	$1/\lambda^2$	<code>exppdf(x, 1/λ)</code> ← mean!
Gaussian N(μ, σ^2)	$(1/\sigma\sqrt{2\pi})e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2	<code>normcdf(x,mu,sigma)</code>

7. KEY MATLAB COMMANDS — GUIDE 2 REFERENCE

Command	Purpose
<code>stem(x, pmf)</code>	Plot discrete PMF (vertical lines)
<code>stairs([x0 x], [0 F])</code>	Plot CDF step function — add 0 at left!
<code>sum(x .* p)</code>	Compute E[X] from PMF vectors x, p
<code>sum((x-EX).^2 .* p)</code>	Compute Var[X] from PMF vectors
<code>poisspdf(k, lambda)</code>	Poisson PMF: $P(X=k)$
<code>poisscdf(k, lambda)</code>	Poisson CDF: $P(X \leq k)$
<code>poissrnd(lambda, r, c)</code>	Generate $r \times c$ Poisson random values
<code>exppdf(x, 1/lambda)</code>	Exponential PDF (pass MEAN not rate!)
<code>exprnd(1/lambda, r, c)</code>	Generate exponential samples (pass MEAN!)
<code>normcdf(x, mu, sigma)</code>	Gaussian CDF: $P(X \leq x)$
<code>normcdf(b,m,s)-normcdf(a,m,s)</code>	$P(a < X < b)$ for Gaussian
<code>randn(r, c)</code>	Standard Normal N(0,1) samples
<code>mu + sigma*randn(r,c)</code>	N(mu, sigma ²) samples
<code>histogram(v, N, "Normalization","pdf")</code>	Density histogram (compare to PDF)
<code>mean(v), var(v), std(v)</code>	Sample mean, variance, std deviation

8. EXAM TIPS & COMMON MISTAKES

- **Poisson: mean = variance.** If asked to find E[X] or Var[X] for a Poisson RV, both equal λ . This is the only standard distribution where this holds.
- **Geometric memoryless trap:** $P(X > 3 | X > 1) \neq P(X > 3)$. Instead $P(X > 3 | X > 1) = P(X > 2) = (1-p)^2$. Subtract the conditioning value from the threshold.
- **Exponential in MATLAB:** `exprnd(mu)` where $\mu = 1/\lambda$ (the MEAN). For $\lambda = 0.2$, pass 5. For $\lambda = 1$, pass 1. Getting this wrong gives completely wrong results.
- **Poisson time scaling:** Always multiply λ by the time window. Rate=5/s over 2 seconds → use $\lambda=10$, not 5.

- **Three inequality bounds:** Markov (needs $X \geq 0$, uses mean) gives loosest bound. Chebyshev (uses mean+variance) is tighter. CLT (for sums of i.i.d.) gives the best estimate.
- **CLT 68% rule:** $P(\mu - \sigma < S_n < \mu + \sigma) \approx 0.6827$. For $S_{100} \sim N(2000, 900)$: interval (1970, 2030) = (2000-30, 2000+30) = $\pm 1\sigma \rightarrow 68.27\%$.
- **Binomial $\rightarrow$ Poisson:** Use Poisson when n is large and p is small. The approximation is excellent when $\lambda = np < 10$ and $n > 100$.
- **CDF plots with stairs:** Always prepend a 0 to the CDF vector and a left-edge x value: `stairs([x_min-1, x_values], [0, cdf_values])`.
- **$E[X]$ from simulation:** `sum(k_values .* pmf_simulated)` — not `mean()` of raw samples. Both work, but the PMF method connects theory to practice.
- **2-engine vs 4-engine plane:** For $p < 0.5$, fly with 2 engines ($P_{\text{crash}2} = p^2 < P_{\text{crash}4} = 4p^3(1-p) + p^4$). The crossover is exactly at $p=0.5$.